

PROBLEMS CLASS 4: SOLUTIONS

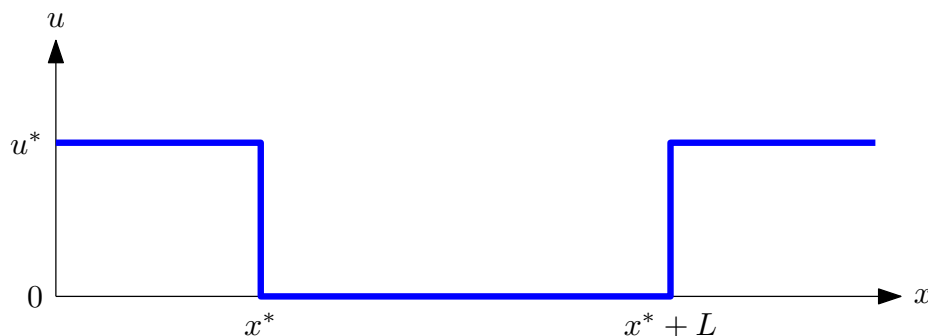
1. Travelling waves in skin repair (Exam 2022, section B)

A model of wound healing in one spatial dimension is given by

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - u \log \left(\frac{u + a}{K} \right), \quad (1)$$

where $u(x, t)$ is the density of healthy tissue at position x and time t ; and where a and K are positive constants, with $0 < a < K$.

- Consider first the homogeneous form of (1). Find the equilibria and determine their stability in this case.
- Describe qualitatively the full model, (1). In what way is it different to that described by the Fisher–Kolmogorov equation?
- We are going to look for travelling wave solutions, i.e. solutions of the form $u(z)$, where $z = x - ct$ and $c > 0$. Given the stability of the equilibria in the homogeneous system, suggest suitable boundary conditions for $u(z)$ at $z = \pm\infty$, explaining your choice.
- Now write (1) for $u(z)$, and by performing a linear stability analysis, determine the stability of the homogeneous equilibria. Given that waves travel at their lowest possible speed, determine the wavespeed, c .
- Consider now a wound which is initially of length L , i.e. as in the diagram below, where u^* is the value of the nonzero homogeneous equilibrium of (1):



Illustrating your answer with sketches of $u(x, t)$ for increasing t , explain what you expect to happen for subsequent times and determine approximately how long it will take for the wound to heal.

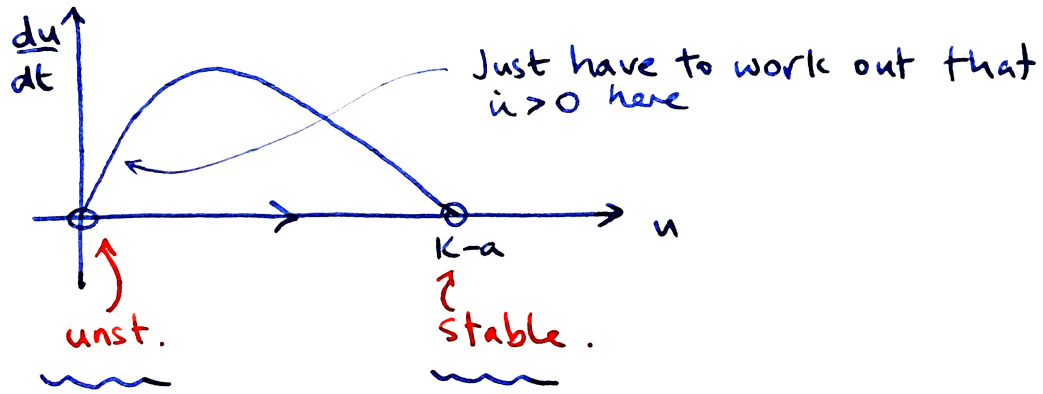
Solution:

- The homogeneous equation is

$$\frac{du}{dt} = -u \log \left(\frac{u + a}{K} \right).$$

The equilibria are at $u = 0$ and $u = K - a$.

Their stability can be determined graphically:



or spotting that for $0 < u \ll 1$, $du/dt > 0$ because $\log(a/K) < 0$ since $a < K$. So $u = 0$ is unstable and since the stability of 1D systems alternates, $u = K - a$ must be stable.

(b) Description of the model:

- diffusion with a growth term, which has a carrying capacity $u = K - a$,
- similar to FK (see graph above) in shape and having a carrying capacity, but FK is logistic, ' $+u(1 - u)$ '.

(c) Argument of the form:

- Expect the stable equilibrium, $u = K - a$, to expand, so the wave will travel away from this.
- A wave of the form $z = x - ct$ moves to the *right*.
- Therefore pick

$$u(-\infty) = K - a, \quad u(\infty) = 0.$$

(d) Substituting in $z = x - ct$, we have

$$\frac{\partial u}{\partial t} = -c \frac{du}{dz} \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = \frac{d^2 u}{dz^2},$$

so

$$-c \frac{du}{dz} = \frac{d^2 u}{dz^2} - u \log \left(\frac{u + a}{K} \right).$$

To perform the linear stability analysis, let $v = du/dz$ then

$$\begin{aligned} \frac{du}{dz} &= v, \\ \frac{dv}{dz} &= -cv + u \log \left(\frac{u + a}{K} \right). \end{aligned}$$

The Jacobian is

$$J = \begin{pmatrix} 0 & 1 \\ \log \left(\frac{u+a}{K} \right) + \frac{u}{u+a} & -c \end{pmatrix},$$

where the bottom-left entry is worked out as

$$\begin{aligned} \frac{d}{du} \left[u \log \left(\frac{u+a}{K} \right) \right] &= \log \left(\frac{u+a}{K} \right) + u \frac{K}{u+a} \frac{1}{K} \\ &= \log \left(\frac{u+a}{K} \right) + \frac{u}{u+a}. \end{aligned}$$

Evaluated at $(0, 0)$, the Jacobian is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ \log \frac{a}{K} & -c \end{pmatrix}.$$

The eigenvalue equation is therefore

$$\begin{aligned} -\lambda(-\lambda - c) - \log \frac{a}{K} &= 0 \\ \implies \lambda^2 + c\lambda - \log \frac{a}{K} &= 0, \end{aligned}$$

and hence

$$\implies \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4 \underbrace{\log \frac{a}{K}}_{<0}} \right).$$

This is therefore a stable node (two negative, real eigenvalues) if

$$c^2 > -4 \log \frac{a}{K}, \quad (2)$$

else it is a spiral (complex eigenvalues).

The Jacobian, evaluated at $(K - a, 0)$, is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ \frac{K-a}{K} & -c \end{pmatrix},$$

and hence

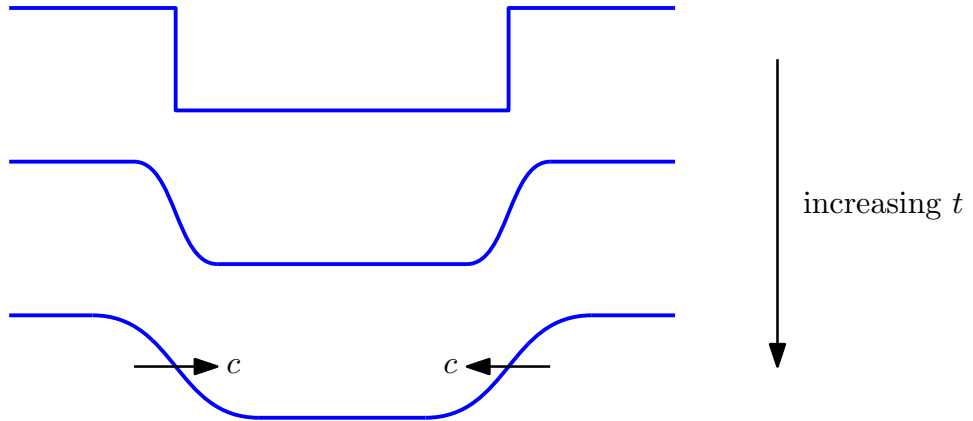
$$\implies \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4 \underbrace{\frac{K-a}{K}}_{>0}} \right).$$

The eigenvalues are real, with different sign. This is therefore a saddle point.

A spiral is not possible, as that would mean negative u , which is unphysical. We therefore conclude eq. (2) is true, and, further, since waves travel at their lowest possible speed,

$$c = 2\sqrt{\log \frac{K}{a}}.$$

(e) Expect $u(x, t)$ to develop like:



with a closure time of approximately

$$\frac{L/2}{c} = \frac{L}{4\sqrt{\log(K/a)}}.$$

2. Disease modelling with vaccination (Exam 2022, section B)

Parliament, 10 November 2020 (Hansard):

Jonathan Ashworth, shadow secretary of state for health and social care: What is the government's current working assumption of the proportion of the population that needs to be vaccinated to establish herd immunity and bring R below 1?

Matt Hancock, secretary of state for health and social care: The honest truth to that question is that we do not know what proportion of the population the vaccination needs to reach in order for it to stop the epidemic.

Consider the following extension of the SIR model for an infectious disease, where susceptible individuals are vaccinated at a rate ν ,

$$\frac{dS}{dt} = \alpha R - \beta IS - \nu S, \quad (3)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (4)$$

$$\frac{dR}{dt} = \gamma I - \alpha R + \nu S,$$

where $\alpha, \beta, \gamma, \nu$ are all positive constants, and $S(t), I(t), R(t) \geq 0$ to be permissible.

- (a) Show that $N = S + I + R$ is constant and use this to eliminate R from (3). Explain why, from now on, we only need to consider the subsystem formed of equations (3) and (4).
- (b) Give an interpretation of the constants α, β, γ, N . What would be suitable units for these constants?
- (c) It is proposed that this could be an appropriate model for the spread of Covid-19. Is this a suitable model for this problem? Give three clear arguments, which can be in favour of this as a suitable model, or against its suitability. Your three arguments do not have to argue the same way.
- (d)
 - (i) Find the values of S and I at the two equilibria of the system.
 - (ii) The disease is eradicated if the disease-free equilibrium is the only possible stable equilibrium and the endemic equilibrium does not exist. Perform a linear stability analysis on both equilibria of the system and show that the disease is eradicated if $R^* < 1$, where you should define the constant R^* .
 - (iii) Define R_0 as the value of R^* in the absence of vaccination (i.e. when $\nu = 0$): in this way, R_0 can be seen as an inherent property of the disease. Rearrange the statement $R^* < 1$ to find the condition on the vaccination rate, ν , for the disease to be eradicated in terms of R_0 .
- (e) You are now asked to present the same argument graphically. Draw a phase portrait of I against S and, by moving the nullclines on your portrait, show how the existence of the endemic equilibrium depends on the vaccination rate, ν , and the value of R_0 . You do not need to explicitly draw trajectories to make this argument.

Solution:

(a) We have that

$$\begin{aligned}\frac{dN}{dt} &= \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} \\ &= (\alpha R - \beta IS - \nu S) + (\beta IS - \gamma I) + (\gamma I - \alpha R + \nu S) \\ &= 0.\end{aligned}$$

Eliminating R , our system becomes

$$\begin{aligned}\frac{dS}{dt} &= \alpha(N - S - I) - \beta IS - \nu S, \\ \frac{dI}{dt} &= \beta IS - \gamma I.\end{aligned}$$

We can lose the equation for dR/dt since dS/dt and dI/dt are independent of R , and because

$$\frac{dR}{dt} = -\frac{dS}{dt} - \frac{dI}{dt}$$

anyway.

(b) Interpretation:

- α : rate at which immunity wears off; any unit that fits the dimension of time^{-1} ,
- β : infection rate; $\text{people}^{-1}\text{time}^{-1}$,
- γ : recovery rate (possibly death rate, but dubious since it is possible to leave the group R); time^{-1} .

(c) Example arguments [students should provide three]:

- no because there is no incubation period (as in an SEIR model),
- no because dead people cannot be reinfected
- no because vaccination doesn't necessarily stop infection, it may just reduce the rate of infection
- no because no spatial dependence
- yes because it captures the general shape of epidemic 'waves'

(d) (i) Equilibria are where

$$0 = \alpha(N - S - I) - \beta IS - \nu S, \quad (5)$$

$$0 = \beta IS - \gamma I. \quad (6)$$

Equation (6) gives us

$$I = 0 \quad \text{or} \quad S = \frac{\gamma}{\beta}.$$

If $I = 0$ then eq. (5) gives us

$$\begin{aligned}0 &= \alpha(N - S) - \nu S \\ \implies \alpha N &= (\alpha + \nu)S \\ \implies S &= \frac{\alpha}{\alpha + \nu}N.\end{aligned}$$

And if $S = \gamma/\beta$ then

$$\begin{aligned}
0 &= \alpha \left(N - \frac{\gamma}{\beta} - I \right) - I\gamma - \frac{\nu\gamma}{\beta} \\
\implies \alpha N - \frac{\alpha\gamma}{\beta} - \frac{\gamma\nu}{\beta} &= I(\alpha + \gamma) \\
\implies \alpha N - (\alpha + \nu) \frac{\gamma}{\beta} &= I(\alpha + \gamma) \\
\implies I &= \frac{\alpha\beta N - (\alpha + \nu)\gamma}{\beta(\alpha + \gamma)}.
\end{aligned}$$

The two equilibria are therefore

$$\left(\frac{\alpha N}{\alpha + \nu}, 0 \right) \quad \text{and} \quad \left(\frac{\gamma}{\beta}, \frac{\alpha\beta N - (\alpha + \nu)\gamma}{\beta(\alpha + \gamma)} \right) =: (S^*, I^*).$$

(ii) The Jacobian is

$$\mathbf{J} = \begin{pmatrix} -\alpha - \beta I - \nu & -\alpha - \beta S \\ \beta I & \beta S - \gamma \end{pmatrix}.$$

Evaluated at $\left(\frac{\alpha N}{\alpha + \nu}, 0 \right)$, it is

$$\begin{pmatrix} -\alpha - \nu & -\alpha - \frac{\beta\alpha N}{\alpha + \nu} \\ 0 & \frac{\beta\alpha N}{\alpha + \nu} - \gamma \end{pmatrix}.$$

The eigenvalues are

$$-\alpha - \nu < 0 \quad \text{and} \quad \frac{\beta\alpha N}{\alpha + \nu} - \gamma.$$

So we can say the eigenvalue at $\left(\frac{\alpha N}{\alpha + \nu}, 0 \right)$ is unstable if $\frac{\beta\alpha N}{\alpha + \nu} > \gamma$.

For the endemic equilibrium, (S^*, I^*) , use the trace and determinant method.

$$\begin{aligned}
\text{tr } \mathbf{J} &= \alpha - \beta I^* - \nu + \beta S^* - \gamma \\
&= -\alpha - \beta I^* - \nu < 0.
\end{aligned}$$

And the determinant, noting that the bottom-right entry of the Jacobian is 0, is

$$\begin{aligned}
\det \mathbf{J} &= 0 - (-\alpha - \gamma)(I^*\beta) \\
&= (\alpha + \gamma)I^*\beta > 0.
\end{aligned} \tag{7}$$

Since $\text{tr } \mathbf{J} < 0$ and $\det \mathbf{J} > 0$, we conclude that (S^*, I^*) is always stable if it exists.

Further, (S^*, I^*) exists if

$$\begin{aligned}
\alpha\beta N - (\alpha + \nu)\gamma &> 0 \\
\implies \left(\frac{\alpha}{\alpha + \nu} \right) \beta N &> \gamma,
\end{aligned}$$

which is the same condition.

The disease is therefore eradicated if

$$R^* := \frac{\alpha}{\alpha + \nu} \frac{\beta N}{\gamma} < 1. \tag{9}$$

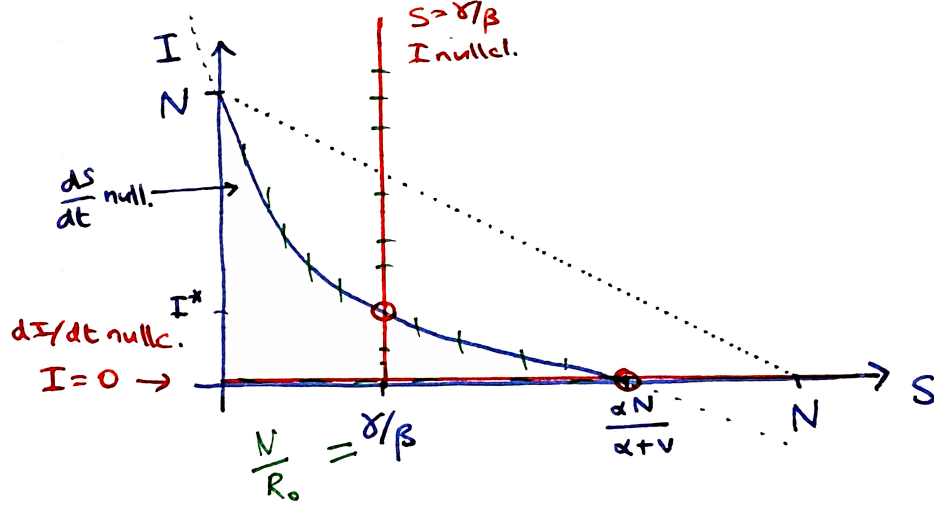
(iii) From eq. (9), we define

$$R_0 = \frac{\beta N}{\gamma},$$

and therefore have

$$\begin{aligned} \frac{\alpha}{\alpha + \nu} R_0 < 1 &\implies \alpha R_0 < \alpha + \nu \\ &\implies \alpha(R_0 - 1) < \nu. \end{aligned}$$

(e) Phase portrait:



The I nullclines are $I = 0$ and $S = \gamma/\beta$.

The S nullcline is where

$$\begin{aligned} 0 &= \alpha N - \alpha S - \alpha I - \beta IS - \nu S \\ \implies I(\alpha + \beta S) &= \alpha(N - S) - \nu S \\ \implies I &= \frac{\alpha N - (\alpha + \nu)S}{\alpha + \beta S}. \end{aligned}$$

This is awkward to plot but notice that

$$I(0) = N \quad \text{and} \quad I\left(\frac{\alpha N}{\alpha + \nu}\right) = 0.$$

You can also notice

- for small S , I looks linear,
- for large S , I looks constant,

which implies the shape in the picture.

As for how the existence of the endemic equilibrium depends on ν and R_0 :

- the equilibrium exists where the S and I nullclines cross,
- the I nullcline is at $\gamma/\beta = N/R_0$,
- changing R_0 shifts the I nullcline left and right,
- changing ν squeezes the S nullcline horizontally,
- it is possible to eradicate the disease by increasing ν or decreasing R_0 .